

Controllability

Can the input steer the state anywhere we want?

Ogata, *Modern Control Engineering*, Ch. 9.

In this tutorial you will:

- test controllability with the `ctrb` rank condition,
- use the PBH eigenvector test to pinpoint an uncontrollable mode, and
- see why controllability is what makes pole placement possible.

Step through with **Ctrl+Enter**, or render a report with **publish**.

A system $\dot{x} = Ax + Bu$ is **completely state controllable** if, for any initial state $x(0)$ and any final state x_1 , there exists an input $u(t)$ that transfers the system from $x(0)$ to x_1 in finite time. For LTI systems this is equivalent to the **controllability matrix**

$$\mathbf{C} = [B \ AB \ A^2B \ \dots \ A^{n-1}B]$$

having full rank n (Kalman's rank condition).

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Example 1: A Controllable System

Inverted-pendulum-like double integrator with cross-coupling:

```
A1 = [0 1; -2 -3];
B1 = [0; 1];
Co1 = ctrb(A1,B1);
fprintf('Example 1: Controllability matrix = \n')
disp(Co1)
fprintf('rank = %d (n = %d) -> %s\n', rank(Co1), size(A1,1), ...
    string(rank(Co1)==size(A1,1)))
```

```
Example 1: Controllability matrix =
     0     1
     1    -3
```

```
rank = 2 (n = 2) -> true
```

Example 2: An Uncontrollable System

Two decoupled subsystems where the input only reaches one of them:

```
A2 = [-1 0; 0 -2];
B2 = [1; 0];
Co2 = ctrb(A2,B2);
fprintf('\nExample 2: Controllability matrix = \n')
disp(Co2)
fprintf('rank = %d (n = %d) -> %s\n', rank(Co2), size(A2,1), ...
        string(rank(Co2)==size(A2,1)))
fprintf('State x2 cannot be influenced by u: it is uncontrollable.\n')
```

Example 2: Controllability matrix =

```
1    -1
0     0
```

```
rank = 1 (n = 2) -> false
State x2 cannot be influenced by u: it is uncontrollable.
```

Geometric Interpretation: Reachable Subspace

The columns of **C** span the **reachable subspace** – the set of states attainable from the origin. For Example 2, this subspace is only the x_1 -axis.

```
fprintf('\nReachable subspace basis (Example 2): column space of Co2\n')
disp(orth(Co2))
```

Reachable subspace basis (Example 2): column space of Co2

```
1.0000
0
```

PBH (Popov-Belevitch-Hautus) Eigenvector Test

An equivalent test: (A, B) is controllable iff $\text{rank} [A - \lambda I \ B] = n$ for every eigenvalue λ of A . This identifies **which** mode is uncontrollable.

```
eigsA2 = eig(A2);
for i = 1:length(eigsA2)
    lam = eigsA2(i);
    PBH = [A2-lam*eye(2) B2];
    fprintf('lambda=%.2f: rank[A-lambda*I B] = %d\n', lam, rank(PBH))
end
```

```
lambda=-2.00: rank[A-lambda*I B] = 1
lambda=-1.00: rank[A-lambda*I B] = 2
```

Controllability of a Three-State System (Mechanical Example)

Two masses connected by a spring, force applied only to mass 1 (Ogata-style coupled mechanical system), augmented with a damped third state to illustrate a 3x3 rank test.

```
A3 = [0 1 0; -2 -1 1; 0 1 -3];
B3 = [0;1;0];
Co3 = ctrb(A3,B3);
fprintf('\nExample 3: rank(Co3) = %d (n=%d)\n', rank(Co3), size(A3,1))
```

Example 3: rank(Co3) = 3 (n=3)

Effect of Controllability on Pole Placement

A controllable pair permits arbitrary closed-loop pole placement via state feedback $u = -Kx$ (see `PolePlacement.m`); an uncontrollable mode cannot have its associated eigenvalue moved by any K .

```
K_test = place(A1,B1,[-3 -4]);
fprintf('\nFeedback gain K placing Example 1 poles at -3,-4: ')
disp(K_test)
fprintf('Closed-loop poles: ')
disp(eig(A1-B1*K_test))

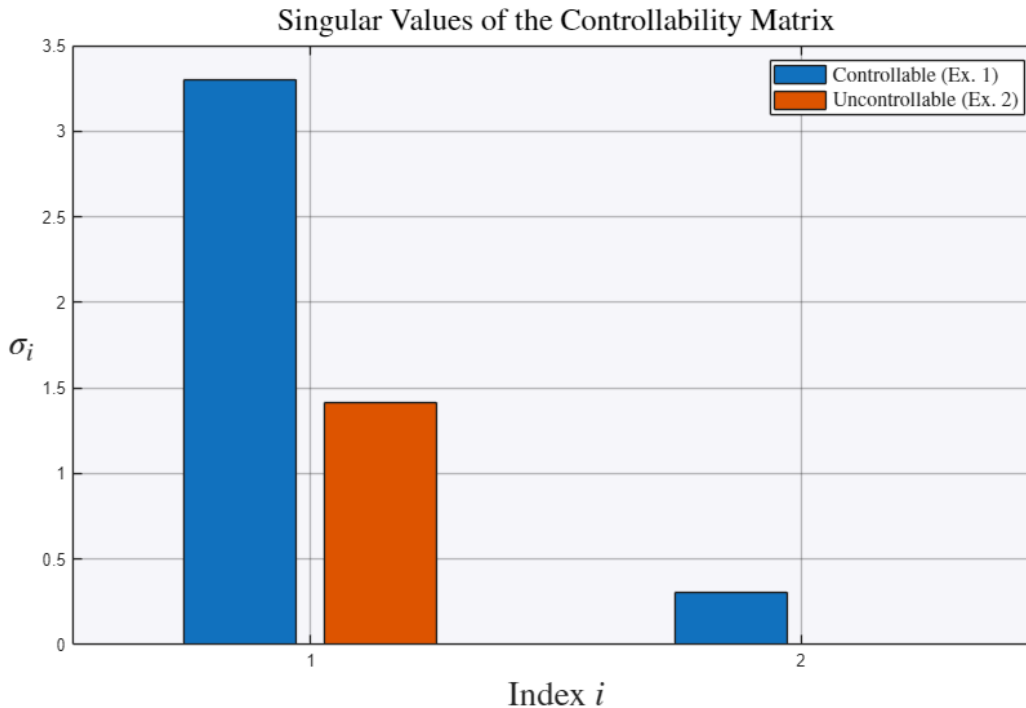
Feedback gain K placing Example 1 poles at -3,-4:    10.0000    4.0000

Closed-loop poles:    -3.0000   -4.0000
```

Visualizing the Rank Condition

The singular values of the controllability matrix make the rank test visual: a controllable pair has all n singular values nonzero, while an uncontrollable pair has a zero singular value flagging the unreachable mode (its bar collapses to zero height below).

```
figure
bar([svd(Co1), svd(Co2)])
grid on
legend('Controllable (Ex. 1)', 'Uncontrollable (Ex. 2)', 'Interpreter', 'latex', 'FontSize', 12)
title('Singular Values of the Controllability Matrix', 'Interpreter', 'latex', 'FontSize', 18)
ylabel('\sigma_i', 'Interpreter', 'latex', 'FontSize', 20)
set(get(gca, 'YLabel'), 'Rotation', 0)
xlabel('Index  $i$ ', 'Interpreter', 'latex', 'FontSize', 20)
```



Try it yourself

- Change B2 to `[1;1]` and notice Example 2 become controllable – the input now reaches both modes.
- Confirm you cannot place the eigenvalue of an uncontrollable mode no matter what gain you choose.